E Proofs for LNSA as a similarity metric

E.1 Invariance to Isotropic Scaling

Lemma E.1 (Invariance to Isotropic scaling in the first coordinate)**.** *Let X and Y be two point clouds of the same size. Let $c \in \mathbf{R}$ and $c \neq 0$, X_c be the point cloud with each point in X scaled by a factor of c, then* $\mathsf{LNSA}^{\mathsf{metric}}(X, Y) = \mathsf{LNSA}^{\mathsf{metric}}(X_c, Y)$.

Proof. Let $X = \{\vec{x}_1, \ldots, \vec{x}_N\}$ and $Y = \{\vec{y}_1, \ldots, \vec{y}_N\}$, then $X_c = \{c\vec{x}_1, \ldots, c\vec{x}_N\}$. We show that $\mathsf{AD}_{X_c}(i, (i_1, \ldots, i_k)) = \mathsf{AD}_X(i, (i_1, \ldots, i_k))$. Note that

$$\mathsf{AD}_{X_c}(i, (i_1, \ldots, i_k)) = -\left(\frac{1}{k}\sum_{j=1}^{k} \log\left(\frac{d(c\vec{x}_i, c\vec{x}_{i_j})}{d(c\vec{x}_i, c\vec{x}_{i_k})}\right)\right)^{-1}.$$

Since, $\frac{d(c\vec{x}_i, c\vec{x}_{i_j})}{d(c\vec{x}_i, c\vec{x}_{i_k})} = \frac{cd(\vec{x}_i, \vec{x}_{i_j})}{cd(\vec{x}_i, \vec{x}_{i_k})} = \frac{d(\vec{x}_i, \vec{x}_{i_j})}{d(\vec{x}_i, \vec{x}_{i_k})}$. Hence,

$$\mathsf{AD}_{X_c}(i, (i_1, \ldots, i_k)) = -\left(\frac{1}{k}\sum_{j=1}^{k} \log\left(\frac{d(\vec{x}_i, \vec{x}_{i_j})}{d(\vec{x}_i, \vec{x}_{i_k})}\right)\right)^{-1}.$$

Hence, $\mathsf{AD}_{X_c}(i, (i_1, \ldots, i_k)) = \mathsf{AD}_X(i, (i_1, \ldots, i_k))$. Hence, we get $\mathsf{lid}(\cdot, X_c) = \mathsf{lid}(\cdot, X)$. Also note that since $kNN_{X_c}(\cdot) = kNN_X(\cdot)$, we have that $\overline{\mathsf{lid}}_{X_c}(\cdot, Y) = \overline{\mathsf{lid}}_X(\cdot, Y)$. Hence, we get $\mathsf{LNSA}^{\mathsf{metric}}(X, Y) = \mathsf{LNSA}^{\mathsf{metric}}(X_c, Y)$. □

By Lemma D.2, we can see that this gives us invariance to isotropic scaling in the second coordinate as well.

Lemma E.2 (Invariance to Isotropic scaling in the second coordinate)**.** *Let X and Y be two point clouds of the same size. Let $c \in \mathbf{R}$ and $c \neq 0$, Y_c be the point cloud with each point in Y scaled by a factor of c, then* $\mathsf{LNSA}^{\mathsf{metric}}(X, Y) = \mathsf{LNSA}^{\mathsf{metric}}(X, Y_c)$.

Combining Lemma E.1 and $Lemma\ E.2$, we get the required lemma.

Lemma E.3 (Invariance to Isotropic scaling)**.** *Let X and Y be two point clouds of the same size. Let $c_1, c_2 \in \mathbf{R}$, $c_1 \neq 0$ and $c_2 \neq 0$, X_{c_1} be the point cloud with each point in X scaled by a factor of c_1 and Y_{c_2} be the point cloud with each point in Y scaled by a factor of c_2, then* $\mathsf{LNSA}^{\mathsf{metric}}(X, Y) = \mathsf{LNSA}^{\mathsf{metric}}(X_{c_1}, Y_{c_2})$.

E.2 Invariance to Orthogonal transformation

Lemma E.4 (Invariance to Orthogonal transformation in the first coordinate)**.** *Let X and Y be two point clouds of the same size. Let U be an orthogonal transformation on the point space of X, X_U be the point cloud with each point in X transformed using U, then* $\mathsf{LNSA}^{\mathsf{metric}}(X, Y) = \mathsf{LNSA}^{\mathsf{metric}}(X_U, Y)$.

Proof. Let $X = \{\vec{x}_1, \ldots, \vec{x}_N\}$ and $Y = \{\vec{y}_1, \ldots, \vec{y}_N\}$, then $X_U = \{U\vec{x}_1, \ldots, U\vec{x}_N\}$. We show that $\mathsf{AD}_{X_U}(i, (i_1, \ldots, i_k)) = \mathsf{AD}_X(i, (i_1, \ldots, i_k))$. Note that

$$\mathsf{AD}_{X_c}(i, (i_1, \ldots, i_k)) = -\left(\frac{1}{k}\sum_{j=1}^{k} \log\left(\frac{d(U\vec{x}_i, U\vec{x}_{i_j})}{d(U\vec{x}_i, U\vec{x}_{i_k})}\right)\right)^{-1}.$$

Since, $\frac{d(U\vec{x}_i, U\vec{x}_{i_j})}{d(U\vec{x}_i, U\vec{x}_{i_k})} = \frac{d(U^T U\vec{x}_i, \vec{x}_{i_j})}{d(U^T U\vec{x}_i, \vec{x}_{i_k})} = \frac{d(\vec{x}_i, \vec{x}_{i_j})}{d(\vec{x}_i, \vec{x}_{i_k})}$. Hence,

$$\mathsf{AD}_{X_c}(i, (i_1, \ldots, i_k)) = -\left(\frac{1}{k}\sum_{j=1}^{k} \log\left(\frac{d(\vec{x}_i, \vec{x}_{i_j})}{d(\vec{x}_i, \vec{x}_{i_k})}\right)\right)^{-1}.$$

Hence, $\mathsf{AD}_{X_U}(i,(i_1,\ldots,i_k)) = \mathsf{AD}_X(i,(i_1,\ldots,i_k))$. Hence, we get $\mathsf{lid}(\cdot,X_U) = \mathsf{lid}(\cdot,X)$. Also note that since $kNN_{X_U}(\cdot) = kNN_X(\cdot)$, we have that $\overline{\mathsf{lid}}_{X_U}(\cdot,Y) = \overline{\mathsf{lid}}_X(\cdot,Y)$. Hence, we get $\mathsf{LNSA}^{\mathsf{metric}}(X,Y) = \mathsf{LNSA}^{\mathsf{metric}}(X_U,Y)$. □

By Lemma D.2, we get invariance to orthogonal transformation in the second coordinate too and combining, we get the required lemma.

Lemma E.5 (Invariance to Orthogonal transformation)**.** *Let X and Y be two point clouds of the same size. Let U_1 be an orthogonal transformation on the point space of X, X_{U_1} be the point cloud with each point in X transformed using U_1. Similarly, let U_2 be an orthogonal transformation on the point space of Y, Y_{U_2} be the point cloud with each point in Y transformed using U_2, then* $\mathsf{LNSA}^{\mathsf{metric}}(X,Y) = \mathsf{LNSA}^{\mathsf{metric}}(X_{U_1},Y_{U_2})$.

E.3 Not Invariant under Invertible Linear Transformation (ILT)

We can easily see this with a counter example. Let $X = \{(1,0,0),(0,1,0),(0,0,1)\}$. Let $Y = \{(1,1,0),(1,0,1),(0,1,1)\}$. Define A as an invertible linear map such that $A(1,0,0) = (2,0,0)$, $A(0,1,0) = (0,1,0)$ and $A(0,0,1) = (0,0,1)$. Let X_A be the point cloud with each point in X transformed using A. Then from some calculations we can see that $\mathsf{LNSA}^{\mathsf{metric}}(X,Y) = 0$ but $\mathsf{LNSA}^{\mathsf{metric}}(X_A,Y) = \frac{1}{12}\left(\log(5)-\log(2)\right)^2$. Hence, we can see that GNSA is not invariant under Invertible Linear Transformations.

F Proofs for LNSA as a Loss Function

F.1 Differentiability of LNSA

To derive the sub-gradient $\frac{\partial \mathsf{LNSA}_Y(X)}{\partial x_i}$, we first define the following notation.

Let

$$f_{j,Y}(X) = \left(\mathsf{lid}(j,Y)^{-1} - \overline{\mathsf{lid}}_Y(j,X)^{-1}\right)^2.$$

$$\text{Using this,} \qquad \mathsf{LNSA}_Y(X) = \frac{1}{N}\sum_{1\le j\le N} f_{j,Y}(X).$$

Hence,

$$\frac{\partial \mathsf{LNSA}_Y(X)}{\partial x_i} = \frac{1}{N}\sum_{1\le j\le N}\frac{\partial f_{j,Y}(X)}{\partial \vec{x}_i}.$$

Note that $\mathsf{lid}(j,Y)^{-1}$ is constant with respect to $\vec{x}_i$, hence,

$$\frac{\partial f_{j,Y}(X)}{\partial \vec{x}_i} = -2\left(\mathsf{lid}(j,Y)^{-1} - \overline{\mathsf{lid}}_Y(j,X)^{-1}\right)\frac{\partial \overline{\mathsf{lid}}_Y(j,X)^{-1}}{\partial \vec{x}_i}.$$

To compute $\frac{\partial \overline{\mathsf{lid}}_Y(j,X)^{-1}}{\partial \vec{x}_i}$, we divide this into the four cases below:

- $i = j$: Let $kNN_Y(i) = (i_1,\cdot,i_k)$, then $\overline{\mathsf{lid}}_Y(i,X)^{-1} = \sum_{l=1}^k \log\left(\frac{d(\vec{x}_i,\vec{x}_{i_l})}{d(\vec{x}_i,\vec{x}_{i_k})}\right)$. Hence,

$$\frac{\partial \overline{\mathsf{lid}}_Y(i,X)^{-1}}{\partial \vec{x}_i} = \sum_{l=1}^k\left(\frac{1}{d(\vec{x}_i,\vec{x}_{i_l})}\frac{\partial d(\vec{x}_i,\vec{x}_{i_l})}{\partial \vec{x}_i} - \frac{1}{d(\vec{x}_i,\vec{x}_{i_k})}\frac{\partial d(\vec{x}_i,\vec{x}_{i_k})}{\partial \vec{x}_i}\right).$$

- $j \in kNN_Y(i)$ **and** j **is not the last element of** $kNN_Y(i)$: Let $kNN_Y(i) = (i_1,\cdot,i_k)$, then $\overline{\mathsf{lid}}_Y(i,X)^{-1} = \sum_{l=1}^k \log\left(\frac{d(\vec{x}_i,\vec{x}_{i_l})}{d(\vec{x}_i,\vec{x}_{i_k})}\right)$. Hence,

$$\frac{\partial \overline{\mathsf{lid}}_Y(i,X)^{-1}}{\partial \vec{x}_i} = \frac{1}{d(\vec{x}_i,\vec{x}_j)}\frac{\partial d(\vec{x}_i,\vec{x}_j)}{\partial \vec{x}_i}.$$

- $j \in kNN_Y(i)$ **and** j **is the last element of** $kNN_Y(i)$: Let $kNN_Y(i) = (i_1, \cdot, i_k)$, then $\overline{\mathsf{lid}}_Y(i, X)^{-1} = \sum_{l=1}^{k} \log\left(\frac{d(\vec{x}_i, \vec{x}_{i_l})}{d(\vec{x}_i, \vec{x}_{i_k})}\right)$. Hence,

$$\frac{\partial \overline{\mathsf{lid}}_Y(i, X)^{-1}}{\partial \vec{x}_i} = (1-k)\left(\frac{1}{d(\vec{x}_i, \vec{x}_j)} \frac{\partial d(\vec{x}_i, \vec{x}_j)}{\partial \vec{x}_i}\right).$$

- $j \notin kNN_Y(i)$ **and** $j \neq i$: Then

$$\frac{\partial \overline{\mathsf{lid}}_Y(i, X)^{-1}}{\partial \vec{x}_i} = 0.$$

Hence, combining the above, we have the sub-gradients of $\mathsf{LNSA}_Y(X)$.

F.2 Continuity

We prove the continuity of $\mathsf{LNSA}^{\mathsf{metric}}$ by showing that $\mathsf{LNSA}^{\mathsf{metric}}$ is a composition of continuous functions (Carothers, 2000, Theorem 5.10). Let

$$f_{i,Y}(X) = \left(\mathsf{lid}(i, Y)^{-1} - \overline{\mathsf{lid}}_Y(i, X)^{-1}\right)^2.$$

$$\text{Using this,} \qquad \mathsf{LNSA}_Y(X) = \frac{1}{N} \sum_{1 \leq i \leq N} f_{i,Y}(X).$$

Since a sum of continuous functions is continuous, all we need to show is that $f_{i,Y}(X)$ is continuous for all $1 \leq i \leq N$. Let $kNN_Y(i) = (i_1, \cdot, i_k)$, then note that

$$f_{i,Y}(X) = \left(\sum_{l=1}^{k}\left(\log\left(\frac{d(\vec{x}_i, \vec{x}_{i_l})}{d(\vec{x}_i, \vec{x}_{i_k})}\right) - \log\left(\frac{d(\vec{y}_i, \vec{y}_{i_l})}{d(\vec{y}_i, \vec{y}_{i_k})}\right)\right)\right)^2,$$

hence can be seen as a composition of $d(\cdot, \cdot)$ and $\log(\cdot)$, along with the algebraic operations of addition, subtraction, multiplication and division. All the above operations are continuous everywhere[5], we get that their composition is also continuous everywhere. Hence, $\mathsf{LNSA}_Y(\cdot)$ is continuous.

G Convergence of subset GNSA

Here, we prove that GNSA adapts well to minibatching and hence can be used as a loss term. In particular, Lemma G.1 proves theoretically that in expectation, GNSA over a minibatch is equal to GNSA of the whole dataset. This supplements the results presented in Figure 2. The experiment is run on the output embeddings of a GCN trained on node classification for the Cora Dataset. We compare the convergence with a batch size of 200 and 500.

Lemma G.1 (Subset GNSA convergence)**.** *Let X, Y be two point clouds of the same size N such that $X = \{x_1, \ldots, x_N\}$ and $Y = \{y_1, \ldots, y_N\}$. Let $\tilde{X}$ be some randomly sampled s sized subsets of X (for some $s \leq N$). Define $\tilde{Y} \subset Y$ as $y_i \in \tilde{Y}$ iff $x_i \in \tilde{X}$. Then the following holds true*

$$\mathop{\mathbb{E}}_{\tilde{X}, \tilde{Y}}\left[\mathsf{GNSA}(\tilde{X}, \tilde{Y})\right] = \mathsf{GNSA}(X, Y).$$

Proof. Let $\max_{x \in X}(d(x, 0)) = D_X$ and $\max_{y \in Y}(d(y, 0)) = D_Y$. Then

$$\mathsf{GNSA}(X, Y) = \frac{1}{N^2} \sum_{1 \leq i,j \leq N} \left|\frac{d(x_i, x_j)}{D_X} - \frac{d(y_i, y_j)}{D_Y}\right|.$$

Define $\mathbb{I}_{\tilde{X}}(x_i)$ as

$$\mathbb{I}_{\tilde{X}}(x_i) = \begin{cases} 1 \text{ if } x_i \in \tilde{X}, \\ 0 \text{ otherwise.} \end{cases}$$

[5]Note that division is not continuous when the denominator goes to zero. Since, the denominators are $(d(\vec{x}_i, \vec{x}_{i_k}))$, as long as all k neighbours of a point are not zero distance away, the denominator does not go to zero.